Run report

2026-10-27T222108Z

Configuration

Parameter	Value	Parameter	Value
batch_size	1,024	n_samples	500,000
dim	1	dataset	gaussian
seed	446,727,034	data_seed	42
hidden_units	64	n_layers	2
n_epochs	100	n_disc_steps	5
lr_g	3.000×10^{-5}	lr_d	1.000×10^{-4}
lambda_dispersion	0.01500	mmd_subsample	16,384
mmd_test	5.333×10^{-3}	best_epoch	98
compile_cache_warm	False		
gaussian_params	dim=1, mu_gen=[0.5], mu_true=[0], cov_gen=[[0.81]], cov_true=[[1]], cov_detector=[[0.25]]		
mmd_sigmas_detector	$0.7470 \times (\frac{1}{2},\ \frac{1}{\sqrt{2}},\ 1,\ \sqrt{2},\ 2)$		
mmd_sigmas_particle	$0.6688 \times (\frac{1}{2},\ \frac{1}{\sqrt{2}},\ 1,\ \sqrt{2},\ 2)$		

Timing

Phase	Seconds	Share (%)	Detail
DECONVOLVE			
data	0.060	0.2	gaussian cache hit – train
train	18.989	74.5	train
transfer	1.828		train
compile	2.237		train
epochs	6.253		train
select	8.051		train
particle_mmd	0.502	2.0	train
save	0.346	1.4	train
evaluate	5.587	21.9	train
total	25.484	100.0	
IBU baseline			
parse_config	0.002	0.0	ibu
data	0.077	1.5	gaussian cache hit – ibu
ibu	0.084	1.6	niter=10 – ibu
evaluate	5.156	96.9	1 variables – ibu
total	5.320	100.0	
OMNIFOLD baseline			
parse_config	0.002	0.0	omnifold
data	0.081	0.0	gaussian cache hit – omnifold
omnifold	368.245	98.5	niter=3, epochs=50 – omnifold
init	3.840		DataLoaders and the two MLPs – omnifold
unfold	305.997		MultiFold.Unfold – omnifold
iter1_step1	67.106		detector-level reweighting – omnifold
iter2_step1	47.089		detector-level reweighting – omnifold
iter3_step1	51.195		detector-level reweighting – omnifold
iter1_step2	43.925		particle-level reweighting – omnifold
iter2_step2	43.403		particle-level reweighting – omnifold
iter3_step2	53.246		particle-level reweighting – omnifold
reweight	1.426		evaluating weights on z_target – omnifold
evaluate	5.478	1.5	1 variables – omnifold
total	373.805	100.0	
Shared by all methods			
parse_config	0.001	0.0	plot
data	0.050	0.2	gaussian cache hit – plot
load	1.561	5.1	plot
plots	29.169	94.8	plot
total	30.781	100.0	

Detector-level metrics

Distance to Data. Improvements are relative to Simulation.

Wasserstein Distance [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	507.8	**18.41**	**96.4**	23.92	95.3	97.76	80.7

JS divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	28.91	**0.5955**	**97.9**	0.7356	97.5	1.782	93.8

VLC divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	111.5	**2.165**	**98.1**	2.709	97.6	6.839	93.9

Sliced Wasserstein, joint [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
All	455.0	**16.49**	**96.4**	21.43	95.3	87.59	80.7

Mean 1D distance over random directions, in Data standard deviations. IBU: product of its per-observable weights.

Particle-level metrics

Distance to Truth. Improvements are relative to Generation.

Wasserstein [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	508.5	42.23	91.7	**23.23**	**95.4**	97.45	80.8

JS divergence [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	37.03	7.828	78.9	**0.7855**	**97.9**	2.417	93.5

VLC divergence [$\times 10^3$]

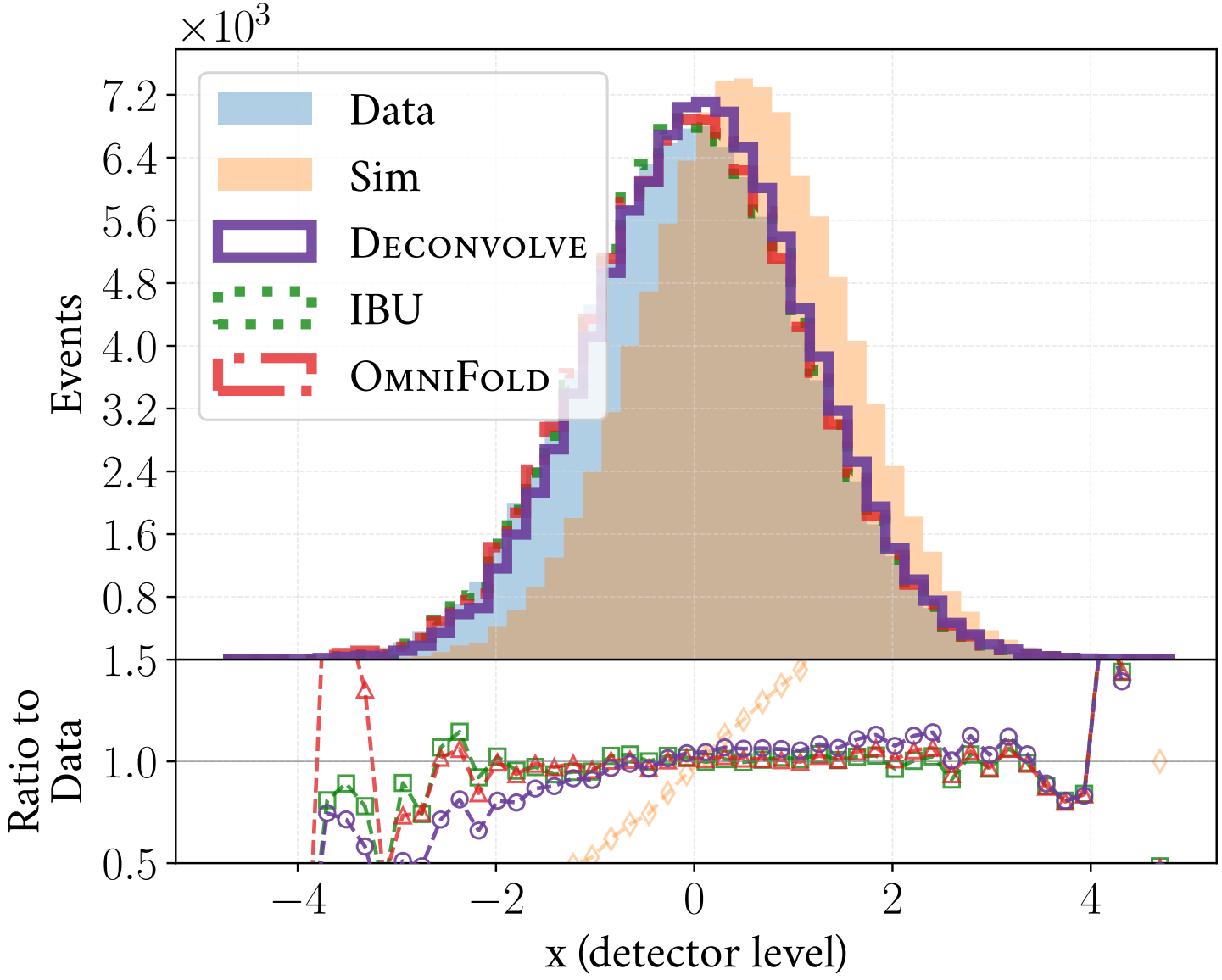
Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	141.3	30.88	78.2	**2.922**	**97.9**	9.293	93.4

Sliced Wasserstein, joint [$\times 10^3$]

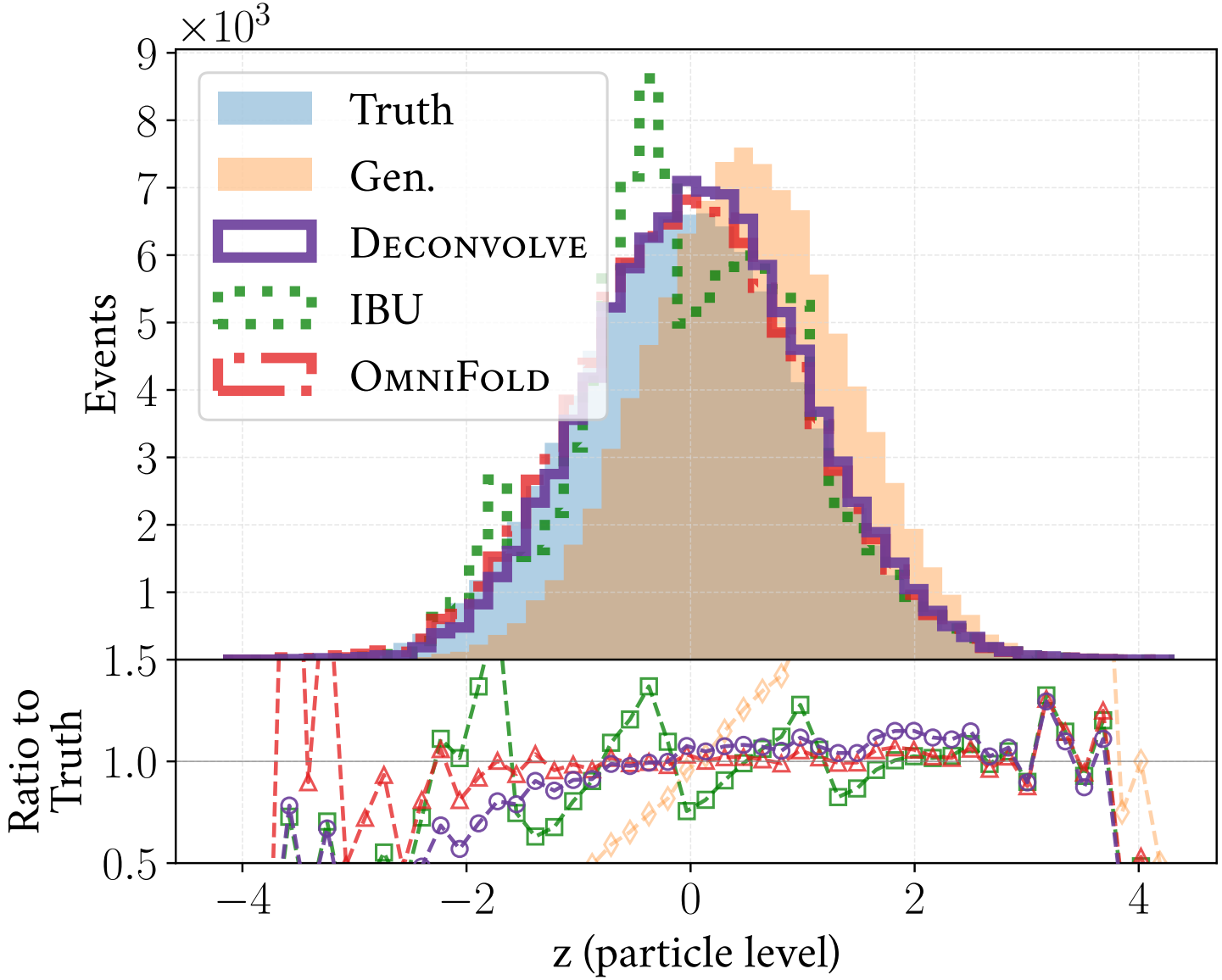
Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
All	508.4	42.23	91.7	**23.23**	**95.4**	97.44	80.8

Mean 1D distance over random directions, in Truth standard deviations. IBU: product of its per-observable weights.

Detector Level



Particle Level



Training diagnostics

Training losses



Selection criteria

